

Lecture Notes on General Relativity

Matthias Blau
Albert Einstein Center for Fundamental Physics
Institut für Theoretische Physik
Universität Bern
CH-3012 Bern, Switzerland

The latest version of these notes is available from

`http://www.blau.itp.unibe.ch/GRLecturenotes.html`

`Last update October 8, 2022`

Contents

0 Introduction **11**

0.1 Prerequisites . . . 12
0.2 Overview . . . 13
0.3 Literature . . . 14
0.4 References and Footnotes . . . 15
0.5 Exercises . . . 16

A: Physics in a Gravitational Field and Tensor Calculus **17**

1 Einstein Equivalence Principle: from Gravity to Geometry **18**

1.1 Motivation: The Einstein Equivalence Principle . . . 18
1.2 Lorentz-Covariant Formulation of Special Relativity (Review) . . . 28
1.3 Accelerated Observers in Minkowski Space and the Rindler Metric . . . 34
1.4 General Coordinate Transformations in Minkowski Space I: Metric . . . 39
1.5 General Coordinate Transformations in Minkowski Space II: Free Particle . . . 41
1.6 General Coordinate Transformations in Minkowski Space III: Lessons . . . 44

2 Metrics, Geometry and Geodesics **48**

2.1 Metrics and Geometry I: Definition and Examples . . . 48
2.2 Metrics and Geometry II: Lorentzian (Pseudo-Riemannian) Metrics . . . 53
2.3 Geodesic Equation from the Extremisation of Proper Time . . . 56
2.4 Geodesic Equation and Coordinate Transformations . . . 60
2.5 Alternative Action Principles for Geodesics . . . 64
2.6 On the Relation between the two Action Principles . . . 66
2.7 Affine and Non-affine Parametrisations . . . 70
2.8 Example: Geodesics in $\mathbb{R}^2$ in Polar Coordinates . . . 72
2.9 Example: Geodesics for Ultrastatic and Direct Product Metrics . . . 75

3 Geodesics and Motion in a Gravitational Field **77**

3.1 Consequences and Uses of the Euler-Lagrange Equations . . . 77
3.2 Conserved Charges and (a first encounter with) Killing Vectors . . . 81
3.3 Newtonian Limit of the Geodesic Equation . . . 83
3.4 Rindler Coordinates Revisited . . . 89
3.5 Gravitational Redshift . . . 93
3.6 Equivalence Principle Revisited: Existence of Locally Inertial Coordinates . . . 101

4 Tensor Algebra **110**

4.1 Principle of General Covariance . . . 110
4.2 Tensors and Tensor Fields . . . 113
4.3 Tensor Algebra . . . 119
4.4 Generally Covariant Integration and Volume Elements . . . 123
4.5 Tensor Densities and Volume Elements . . . 126
4.6 Towards a Coordinate-Independent Interpretation of Tensors . . . 129
4.7 Multilinear Algebra and Tensors . . . 131
4.8 Vielbeins and Orthonormal Frames . . . 135

4.9 Epilogue: Indices? Indices! . . . 141

5 Tensor Analysis (Generally Covariant Differentiation) 146
5.1 Covariant Derivative for Vector Fields . . . 146
5.2 Extension of the Covariant Derivative to Other Tensor Fields . . . 149
5.3 Main Properties of the Covariant Derivative . . . 151
5.4 Uniqueness of the Levi-Civita Connection (Christoffel symbols) . . . 153
5.5 Tensor Analysis: Some Special Cases . . . 154
5.6 Appendix: A Formula for the Variation of the Determinant . . . 160
5.7 Covariant Differentiation Along a Curve . . . 162
5.8 Parallel Transport and Geodesics . . . 163
5.9 Example: Parallel Transport on the 2-Sphere . . . 164
5.10 Fermi-Walker Parallel Transport . . . 167
5.11 Epilogue: Manifolds? Think Globally, Act Locally! . . . 170

6 Physics in a Gravitational Field and Minimal Coupling 175
6.1 Principle (or Algorithm) of Minimal Coupling . . . 175
6.2 Particle Mechanics in a Gravitational Field Revisited . . . 176
6.3 Klein-Gordon Scalar Field in a Gravitational Field . . . 177
6.4 Interlude: General Covariance in Minkowski Space? . . . 178
6.5 Lorentz-Covariant Formulation of Maxwell Theory (Review) . . . 180
6.6 Maxwell Theory in a Gravitational Field . . . 182
6.7 Minimal Coupling and (quasi-)Topological Couplings . . . 186
6.8 Conserved Charges from Covariantly Conserved Currents . . . 188

7 Energy-Momentum Tensor I: Basics 190
7.1 Introduction . . . 190
7.2 Perfect Fluid Energy-Momentum Tensor in Special Relativity . . . 190
7.3 Noether Energy-Momentum Tensor in Special Relativity (Review) . . . 193
7.4 Synopsis of the Belinfante Improvement Procedure (Review) . . . 197
7.5 Energy-Momentum Tensor from Minimal Coupling? . . . 199
7.6 Covariant Energy-Momentum Tensor: the Source of Gravity . . . 203
7.7 On the Energy-Momentum Tensor for Weyl-invariant Actions . . . 207
7.8 Klein-Gordon Scalar Field in (1+1) Minkowski and Rindler Space . . . 209
7.9 Conserved Currents from the Energy-Momentum Tensor? . . . 212

8 Curvature I: The Riemann Curvature Tensor 215
8.1 Curvature: Preliminary Remarks . . . 215
8.2 Riemann Tensor from the Commutator of Covariant Derivatives . . . 216
8.3 Symmetries and Algebraic Properties of the Riemann Tensor . . . 218
8.4 Tidal Forces: Influence of Curvature on Particle Trajectories . . . 222
8.5 Contractions of the Riemann Tensor: Ricci Tensor and Ricci Scalar . . . 224
8.6 Example: Curvature Tensor of the 2-Sphere . . . 227
8.7 More Examples: Curvature Tensor and Polar/Spherical Coordinates . . . 228
8.8 Bianchi Identities and the Einstein Tensor . . . 232
8.9 Riemann Normal Coordinates Revisited . . . 234
8.10 Principle of Minimal Coupling Revisited . . . 236

B: General Relativity and Geometry **238**

9 Lie Derivative, Symmetries and Killing Vectors **239**
9.1 Symmetries of a Metric (Isometries): Preliminary Remarks . . . 239
9.2 Lie Derivative for Scalars . . . 240
9.3 Lie Derivative for Vector Fields . . . 241
9.4 Lie Derivative for other Tensor Fields . . . 244
9.5 Lie Derivative of the Metric and Killing Vectors . . . 246
9.6 Lie Derivative for Tensor Densities . . . 250

10 Killing Vectors, Symmetries and Conserved Charges **252**
10.1 Killing Vectors and Conserved Charges . . . 252
10.2 Conformal Killing Vectors and Conserved Charges . . . 253
10.3 Conformal Group and Conformal Algebra of Minkowski Space . . . 255
10.4 Homotheties and Conserved Charges . . . 257
10.5 Conserved Charges from Killing Tensors and Killing-Yano Tensors . . . 259

11 Curvature II: Geometry and Curvature **262**
11.1 Intrinsic Geometry, Curvature and Parallel Transport . . . 262
11.2 Vanishing Riemann Tensor and Existence of Flat Coordinates . . . 266
11.3 Curvature of Surfaces: Euler, Gauss(-Bonnet) and Liouville . . . 268
11.4 The Weyl Tensor and its Uses . . . 274
11.5 Generalisations: Torsion and Non-Metricity . . . 278

12 Curvature III: Curvature and Geodesic Congruences **283**
12.1 Covariant Derivation of the Geodesic Deviation Equation . . . 283
12.2 Raychaudhuri Equation for Timelike Geodesic Congruences . . . 285
12.3 Transverse Null Geodesic Deviation Equation . . . 291
12.4 Raychaudhuri Equation for Affine Null Geodesic Congruences . . . 296
12.5 Raychaudhuri Equation for Non-affinely Parametrised Null Geodesics . . . 300
12.6 Expansions and Inaffinities of Radial Null Congruences . . . 302

13 Curvature IV: Curvature and Killing Vectors **305**
13.1 Useful Identities Relating Curvature and Killing Vectors . . . 305
13.2 Killing Vectors form a Lie algebra . . . 307
13.3 On the Isometry Algebra of a Compact Riemannian Space . . . 308
13.4 Invariance of the Curvature along Killing Directions . . . 309
13.5 Calculating Killing Components of the Ricci Tensor . . . 310
13.6 Killing Vectors as Solutions to the Maxwell Equations . . . 311
13.7 Killing Vectors and Komar Currents . . . 312

14 Curvature V: Maximal Symmetry and Constant Curvature **314**
14.1 Homogeneous, Isotropic and Maximally Symmetric Spaces . . . 314
14.2 Curvature Tensor of a Maximally Symmetric Space . . . 317
14.3 Maximally Symmetric Metrics I: Solving the Constant Curvature Conditions . . . 318
14.4 Maximally Symmetric Metrics II: Embeddings . . . 321

15 Hypersurfaces I: Basics **323**

15.1 Basic Definitions: Embeddings and Embedded Hypersurfaces . . . 323
15.2 Embeddings: Tangent and Normal Vectors and the Induced Metric . . . 327
15.3 Embeddings and Pull-Backs . . . 330
15.4 Embedded Hypersurfaces and Normal Vectors . . . 334
15.5 Hypersurface Orthogonality and Frobenius Integrability . . . 336

16 Hypersurfaces II: Intrinsic Geometry of non-Null Hypersurfaces 338
16.1 Projectors for non-Null Hypersurfaces and the Induced Metric . . . 338
16.2 Intrinsic = Projected Covariant Differentiation . . . 341
16.3 Integration on non-Null Hypersurfaces and the Gauss Theorem . . . 342
16.4 Spacelike Hypersurfaces and Stationary vs Static Metrics . . . 344

17 Hypersurfaces III: Intrinsic Geometry of Null Hypersurfaces 351
17.1 Null Hypersurfaces . . . 351
17.2 Null Hypersurfaces and their Null Geodesic Generators . . . 353
17.3 Adapted Coordinates and Induced Metric for Null Hypersurfaces . . . 357
17.4 Projectors for Null Hypersurfaces . . . 358

18 Hypersurfaces IV: Extrinsic Geometry of non-Null Hypersurfaces 361
18.1 Introduction: Intrinsic vs Extrinsic Geometry . . . 361
18.2 Extrinsic Curvature Tensor . . . 364
18.3 Extrinsic Curvature and the Normal Components of the Connection . . . 368
18.4 Gauss-Codazzi Equations . . . 369

C: Dynamics of the Gravitational Field 373

19 The Einstein Equations 374
19.1 Heuristics . . . 374
19.2 More Systematic Approach . . . 375
19.3 Newtonian Weak-Field Limit . . . 377
19.4 Einstein Equations . . . 379
19.5 Cosmological Constant . . . 382
19.6 Weyl Tensor and the Propagation of Gravity . . . 384
19.7 General Covariance and Significance of the Bianchi Identities . . . 385

20 Einstein Equations from an Action Principle 388
20.1 Einstein-Hilbert Action . . . 388
20.2 Appendix: A Formula for the Variation of the Ricci Tensor . . . 391
20.3 Matter Action and the Covariant Energy-Momentum Tensor . . . 393
20.4 Einstein Action . . . 395
20.5 Gibbons-Hawking-York Boundary Term . . . 397
20.6 General Covariance and Noether Identities . . . 401
20.7 First Order Form of the Action, Torsion and the Palatini Principle . . . 407

21 Hamiltonian Formulation of General Relativity 414
21.1 General Covariance and Constraints . . . 415
21.2 Gauss-Codazzi Action and the Gibbons-Hawking-York Boundary Term . . . 416
21.3 ADM Decomposition of the Metric (ADM Variables) . . . 418

21.4 ADM Action and the DeWitt Metric . . . 421
21.5 Synopsis of the Canonical Formulation of Maxwell Theory . . . 422
21.6 Back to Gravity: Conjugate Momenta and Primary Constraints . . . 425
21.7 Legendre Transform and ADM Hamiltonian . . . 427
21.8 Secondary Constraints: the Hamiltonian and Momentum Constraints . . . 429
21.9 Properties and Significance of the Constraints . . . 431
21.10 Boundary Terms in the ADM Action and Hamiltonian . . . 436
21.11 Alternative Derivation of the Hamiltonian Boundary Terms . . . 440
21.12 Significance of the Hamiltonian Boundary Terms: ADM Energy . . . 441

22 Energy-Momentum Tensor II: Selected Topics 446
22.1 Energy Conditions . . . 446
22.2 Canonical vs Covariant Energy-Momentum Tensor . . . 454
22.3 Energy-Momentum Tensor of a Conformally Coupled Scalar Field . . . 459
22.4 Remarks on Dilatations and the Callan-Coleman-Jackiw Tensor . . . 466
22.5 Energy-Momentum Tensor and (quasi-)Topological Couplings . . . 473
22.6 Comments on Gravitational Energy . . . 477

23 Linearised Gravity and Gravitational Waves 484
23.1 Preliminary Remarks . . . 484
23.2 Linearised Einstein Equations . . . 484
23.3 Newtonian Limit Revisited . . . 487
23.4 ADM and Komar Energies of an Isolated System . . . 488
23.5 Wave Equations and Gauge Conditions in Maxwell Theory . . . 490
23.6 Linearised Gravity: Gauge Invariance and Coordinate Choices . . . 492
23.7 Wave Equation . . . 494
23.8 Polarisation Tensor and the Metric of a Gravitational Wave . . . 495
23.9 Physical Effects of Gravitational Waves . . . 498
23.10 Brief Comments on Production and Energy of Gravitational Waves . . . 500
23.11 Even Briefer Comments on Detection of Gravitational Waves . . . 503

D: General Relativity and the Solar System 505

24 Einstein Equations and Spherical Symmetry 506
24.1 Introduction . . . 506
24.2 Static Spherically Symmetric Metrics . . . 506
24.3 Solving the Einstein Equations: the Schwarzschild Metric . . . 510
24.4 Schwarzschild Coordinates and Schwarzschild Radius . . . 514
24.5 Measuring Length and Time in the Schwarzschild Metric . . . 516
24.6 Einstein Equations for Spherical Symmetry and Birkhoff's Theorem . . . 518
24.7 Interior Solution for a Static Star and the TOV Equation . . . 524
24.8 ADM and Komar Energies of the Schwarzschild Solution . . . 531

25 Particle and Photon Orbits in the Schwarzschild Geometry 535
25.1 Schwarzschild Metric: Summary of its Key Properties . . . 535
25.2 Symmetries and Effective Potential for the Kepler Problem . . . 538
25.3 Symmetries and Effective Potential for Schwarzschild Geodesics . . . 539

25.4 Caveat: Effective Potential versus Routhian . . . 545
25.5 Equation for the Shape of the Orbit . . . 546
25.6 Timelike Geodesics . . . 548
25.7 Some Comments on Circular Timelike Orbits . . . 550
25.8 Anomalous Precession of the Perihelia of the Planetary Orbits . . . 552
25.9 Null Geodesics . . . 555
25.10 Some Comments on Circular Null Orbits (The Photon Sphere) . . . 557
25.11 Bending of Light by a Star: 3 Derivations . . . 558
25.12 A Unified Description in terms of the Runge-Lenz Vector . . . 564

E: Black Holes **568**

26 Black Holes I: Approaching the Schwarzschild Radius r_s **569**
26.1 Preliminary Considerations . . . 569
26.2 Static Observers . . . 571
26.3 Vertical Free Fall . . . 573
26.4 Vertical Free Fall as seen by a Distant Observer . . . 574
26.5 Gravitational Redshift in the Schwarzschild Geometry . . . 576
26.6 Geometry near r_s and Minkowski Space in Rindler Coordinates . . . 580
26.7 Lightcones and Tortoise Coordinates . . . 584
26.8 Massless Klein-Gordon Scalar Field in the Schwarzschild Geometry . . . 587

27 Black Holes II: the Schwarzschild Black Hole **590**
27.1 Preliminary Remarks . . . 590
27.2 Crossing r_s with Painlevé-Gullstrand Coordinates . . . 591
27.3 Lemaître and Novikov Coordinates . . . 597
27.4 Eddington-Finkelstein Coordinates I: Regularity of the Metric . . . 603
27.5 Eddington-Finkelstein Coordinates II: Event Horizons and Black Holes . . . 608
27.6 Eddington Time Coordinate and Kerr-Schild Form of the Metric . . . 613
27.7 Kruskal-Szekeres Coordinates . . . 616
27.8 Maximal Extension of Schwarzschild: the Kruskal Diagram . . . 620
27.9 Properties of the Asymptotically Timelike Killing Vector . . . 624
27.10 First Encounter with Killing Horizons and Surface Gravity . . . 625
27.11 From Eddington-Finkelstein to Israel(-Klösch-Strobl) Coordinates . . . 629
27.12 Appendix: Summary of Schwarzschild Coordinate Systems . . . 634

28 Interlude: Carter-Penrose Conformal Diagrams **635**
28.1 Introduction . . . 635
28.2 Causal Structure and Conformal Rescalings of the Metric . . . 636
28.3 Penrose Diagram for (3+1) Minkowski Space . . . 638
28.4 Penrose Diagram for (1+1) Minkowski Space and Rindler Observers . . . 644
28.5 Penrose Diagram for Schwarzschild (Kruskal-Szekeres) . . . 645
28.6 Penrose Diagrams: List of Other Examples . . . 649

29 Black Holes III: Simple Models of Gravitational Collapse **652**
29.1 Collapse of a Shell of Radiation . . . 652
29.2 Some Comments on Null Shells and Adapted Coordinates . . . 656

29.3 Qualitative Aspects of the Gravitational Collapse of a Star . . . 658
29.4 Oppenheimer-Snyder Set-Up: Geometry and Matter Content . . . 664
29.5 $k = 0$ Collapse and Painlevé-Gullstrand Coordinates . . . 667
29.6 Synopsis of the Oppenheimer-Snyder Construction . . . 670
29.7 Continuity of the Normal Derivatives of the Metric . . . 671
29.8 $k = 1$ Collapse and Comoving Coordinates . . . 674

30 Black Holes IV: Other Black Hole Solutions (a brief overview) 677
30.1 Kerr-Newman Family of 4-dimensional Black Holes . . . 677
30.2 Other 4-dimensional Solutions . . . 680
30.3 Higher-dimensional Solutions . . . 683

31 Black Holes V: the Reissner-Nordstrøm Solution 686
31.1 Derivation of the Reissner-Nordstrøm Metric . . . 686
31.2 Basic Properties of the Naked Singularity Solution with $m^2 - q^2 < 0$ . . . 689
31.3 Basic Properties of the Extremal Solution with $m^2 - q^2 = 0$ . . . 690
31.4 Peculiar Properties of the Extremal Solution with $m^2 - q^2 = 0$ . . . 691
31.5 Basic Properties of the Non-extremal Solution with $m^2 - q^2 > 0$ . . . 694
31.6 Motion of a Charged Particle: the Effective Potential . . . 699
31.7 Eddington-Finkelstein Coordinates: General Considerations . . . 705
31.8 Eddington-Finkelstein Coordinates: the Reissner-Nordstrøm Metric . . . 706
31.9 Kruskal-Szekeres Coordinates: General Considerations . . . 709
31.10 Kruskal-Szekeres Coordinates: the Reissner-Nordstrøm Metric . . . 711

32 Black Holes VI: Horizons 714
32.1 Introduction . . . 714
32.2 The many Facets of the Schwarzschild Radius . . . 715
32.3 Kerr Metric: Ergosphere vs Killing Horizon and Event Horizon . . . 719
32.4 Event Horizon . . . 725
32.5 Killing Horizons as Event Horizons of Stationary Black Holes . . . 728
32.6 Killing Horizons and Surface Gravity . . . 731
32.7 Killing Horizons, Event Horizons and their Null Congruences . . . 735
32.8 Vaidya Metrics: (Marginally) Trapped Surfaces . . . 738
32.9 Vaidya (and beyond): Marginally Trapped Tubes and Horizons . . . 744
32.10 Vaidya Metrics: Apparent Horizon vs Event Horizon . . . 748
32.11 Example: Horizons in the Collapsing Thin Light Shell Geometry . . . 750
32.12 Example: Horizons in Oppenheimer-Snyder Collapse . . . 752
32.13 Concluding Comments: Trapped Regions vs Event Horizons . . . 756

F: Cosmology 760

33 Cosmology I: Basics 761
33.1 Preliminary Remarks . . . 761
33.2 Fundamental Assumption: The Cosmological Principle . . . 762
33.3 Fundamental Observations I: Olbers' Paradox . . . 763
33.4 Fundamental Observations II: The Hubble(-Lemaître) Expansion . . . 765

34 Cosmology II:Geometry and Physics of Robertson-Walker Metrics **767**
34.1 Mathematical Model: the Robertson-Walker Metric . . . 767
34.2 Timelike Geodesics and Comoving Observers . . . 770
34.3 Velocity - Distance Relation, Recessional Velocities and the Hubble Sphere . . . 772
34.4 Painlevé-Gullstrand-like Coordinates for Comoving Observers . . . 776
34.5 Conformal Time η . . . 777
34.6 Conformal Flatness of Robertson-Walker Metrics . . . 778
34.7 Area Measurements and Number Counts . . . 780
34.8 Cosmological Redshift . . . 782
34.9 Redshift - Distance Relation (Hubble's Law) . . . 785
34.10 Klein-Gordon Scalar Field in a Cosmological Background . . . 788
34.11 Comments on Cosmic Expansion as "Expansion of Space" . . . 792

35 Cosmology III:Friedmann-Lemaître-Robertson-Walker Cosmology **795**
35.1 Curvature and Einstein Tensor of the Robertson-Walker Metric . . . 795
35.2 Matter Content: A Perfect Fluid . . . 798
35.3 Appendix: Space-Time Tensors with Maximal Spatial Symmetry . . . 803
35.4 Conservation Laws for Perfect Fluids . . . 805
35.5 Conservation Laws and Comoving Congruences . . . 806
35.6 Conservation Laws for Specific Equations of State . . . 809
35.7 Einstein and Friedmann-Lemaître Equations . . . 811
35.8 Lagrangian Formulation of the Friedmann Equations . . . 814
35.9 Hamiltonian Formulation of the Friedmann Equations . . . 820

36 Cosmology IV: Qualitative Analysis **823**
36.1 The Past I: Big Bang (Existence of an Initial Singularity) . . . 823
36.2 The Past II: Age of the Universe . . . 824
36.3 The Future: Long Term Behaviour . . . 824
36.4 Different Eras of the Cosmological Expansion . . . 827
36.5 Density Parameters and the Critical Density . . . 829
36.6 Causal Structure and Penrose Diagrams for $k = 0$ . . . 831
36.7 Lightrays and Cosmological Particle and Event Horizons . . . 833
36.8 Lightrays and the Hubble Sphere . . . 838
36.9 Hubble "Horizon": Is our Universe (like) a Black Hole? . . . 842

37 Cosmology V: Some Exact Solutions **847**
37.1 Milne Universe . . . 847
37.2 Einstein Static Universe . . . 850
37.3 Matter Dominated Era . . . 851
37.4 Radiation Dominated Era . . . 854
37.5 Cosmological Constant Dominated Era: (Anti-) de Sitter Space . . . 856

38 Cosmology VI: The Universe Today - Insights and Puzzles **860**
38.1 Λ-CDM Cold Dark Matter + Dark Energy Model . . . 860
38.2 Exact Λ-CDM Solution . . . 863
38.3 Flatness and Horizon Problems . . . 864
38.4 Aspects of the Cosmological Constant Problem(s) . . . 870

G: Varia **879**

39 de Sitter and anti-de Sitter Space **880**
39.1 Embeddings, Isometries and Coset Space Structure . . . 880
39.2 Some Coordinate Systems for de Sitter space . . . 883
39.2.1 Global Coordinates . . . 884
39.2.2 Conformal Time, Penrose Diagrams and Horizons . . . 884
39.2.3 Hyperbolic Slicing Coordinates . . . 886
39.2.4 de Sitter Slicing Coordinates . . . 887
39.2.5 Planar Coordinates . . . 888
39.2.6 Static Coordinates . . . 889
39.2.7 Eddington-Finkelstein and Kruskal-Szekeres Coordinates . . . 891
39.2.8 Interlude on (A)dS Schwarzschild . . . 892
39.2.9 Painlevé-Gullstrand-like Coordinates . . . 893
39.3 Some Coordinate Systems for anti-de Sitter space . . . 894
39.3.1 Global (and Static) Coordinates . . . 895
39.3.2 Conformal Coordinates, Conformal Boundary and Penrose Diagrams . . . 896
39.3.3 Isotropic (Spatially Conformally Flat) Coordinates . . . 899
39.3.4 Cosmological (Hyperbolic Slicing) Coordinates . . . 900
39.3.5 de Sitter Slicing Coordinates . . . 901
39.3.6 anti-de Sitter Slicing Coordinates . . . 901
39.3.7 Poincaré Coordinates . . . 902
39.3.8 Plane Wave AdS Coordinates . . . 904
39.3.9 Codimension-2 Hyperbolic Slicing Coordinates . . . 906
39.3.10 Painlevé-Gullstrand-like Coordinates? . . . 907
39.4 Warped Products, Cones, and Maximal Symmetry . . . 909

40 Vaidya Metrics I: Bondi Gauge and Radiation Fields **915**
40.1 Introduction: Ingoing and Outgoing Vaidya Metrics . . . 915
40.2 Interpretation(s) of the Vaidya Mass Functions $m(v)$ and $m(u)$ . . . 918
40.3 Einstein Equations in the Bondi Gauge (Radiative Coordinates) . . . 920
40.4 Description of In- and Outgoing Pure Radiation Fields . . . 923
40.5 Vaidya Metrics in the Schwarzschild Gauge . . . 926

41 Vaidya Metrics II: Radial Null and Timelike Geodesics **930**
41.1 Radial Null Geodesics for Ingoing Vaidya . . . 930
41.2 Radial Null Geodesics for Outgoing Vaidya . . . 933
41.3 Gravitational Redshift for Outgoing Vaidya . . . 934
41.4 Radial Timelike Geodesics for Outgoing Vaidya . . . 936
41.5 Future Incompletetness of Outgoing Eddington-Finkelstein Coordinates . . . 938
41.6 Infinite Gravitational Redshift and Future Incompleteness . . . 939
41.7 Some Comments on Future Extensions of Outgoing Vaidya . . . 941

42 Vaidya Metrics III: Linear Mass $m(v) = \mu v$ (a case study) **944**
42.1 Outgoing Lightrays for $m(v) = \mu v$: Derivation 1 . . . 944
42.2 Some Comments on Homotheties, Geodesics and Wronskians . . . 948
42.3 Event vs Apparent Horizons for $m(v) = \mu v$: Overview . . . 950

42.4 Null Geodesics, Horizons and Singularities for $\mu = 1/16$ . . . 954
42.5 Linear $\mu = 1/16$ mass Vaidya glued to Schwarzschild . . . 956
42.6 Appendix: Outgoing Lightrays for $m(v) = \mu v$: Derivation 2 . . . 959

43 Exact Wave-like Solutions of the Einstein Equations 963
43.1 Plane Waves in Rosen Coordinates: Heuristics . . . 963
43.2 From pp-waves to plane waves in Brinkmann coordinates . . . 964
43.3 Geodesics, Light-Cone Gauge and Harmonic Oscillators . . . 967
43.4 Curvature and Singularities of Plane Waves . . . 969
43.5 From Rosen to Brinkmann coordinates (and back) . . . 971
43.6 More on Rosen Coordinates . . . 973
43.7 Heisenberg Isometry Algebra of a Generic Plane Wave . . . 975
43.8 Plane Waves with more Isometries . . . 976

44 Kaluza-Klein Theory 979
44.1 Motivation: Gravity and Gauge Theory . . . 979
44.2 The Kaluza-Klein Miracle: History and Overview . . . 981
44.3 Origin of Gauge Invariance . . . 984
44.4 Motion of Charged Particles in 4 Dimensions from 5-Dimensional Geodesics . . . 986
44.5 First Problems: The Equations of Motion . . . 987
44.6 Masses and Charges from Scalar Fields in 5 Dimenions . . . 988
44.7 Kinematics of Dimensional Reduction . . . 990
44.8 Kaluza-Klein Ansatz Revisited . . . 991
44.9 Non-Abelian Generalisation and Outlook . . . 993